Multiple Regression

Overall relationships and partial relationships

Foundations of Regression and Predictive Modeling

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Multiple regression

So far, we have mostly asked:

How does y vary with one x?

Now we ask a more grown-up version:

How does y vary with one x, after adjusting for another x?

That phrase, **after adjusting**, is the whole game.

The case

The SDS undergraduate capstone course has students work in teams on semester-long data products.

- Every team uses GitHub.
- Every team submits its AI-assistant logs as part of the assignment.

So for every merged pull request across all teams, we can construct:

- `HoursToMerge`: hours from opening the PR to merge,
- `AIPrompts`: number of AI-assistant prompts linked to the PR,
- `LinesChanged`: total lines added plus deleted.

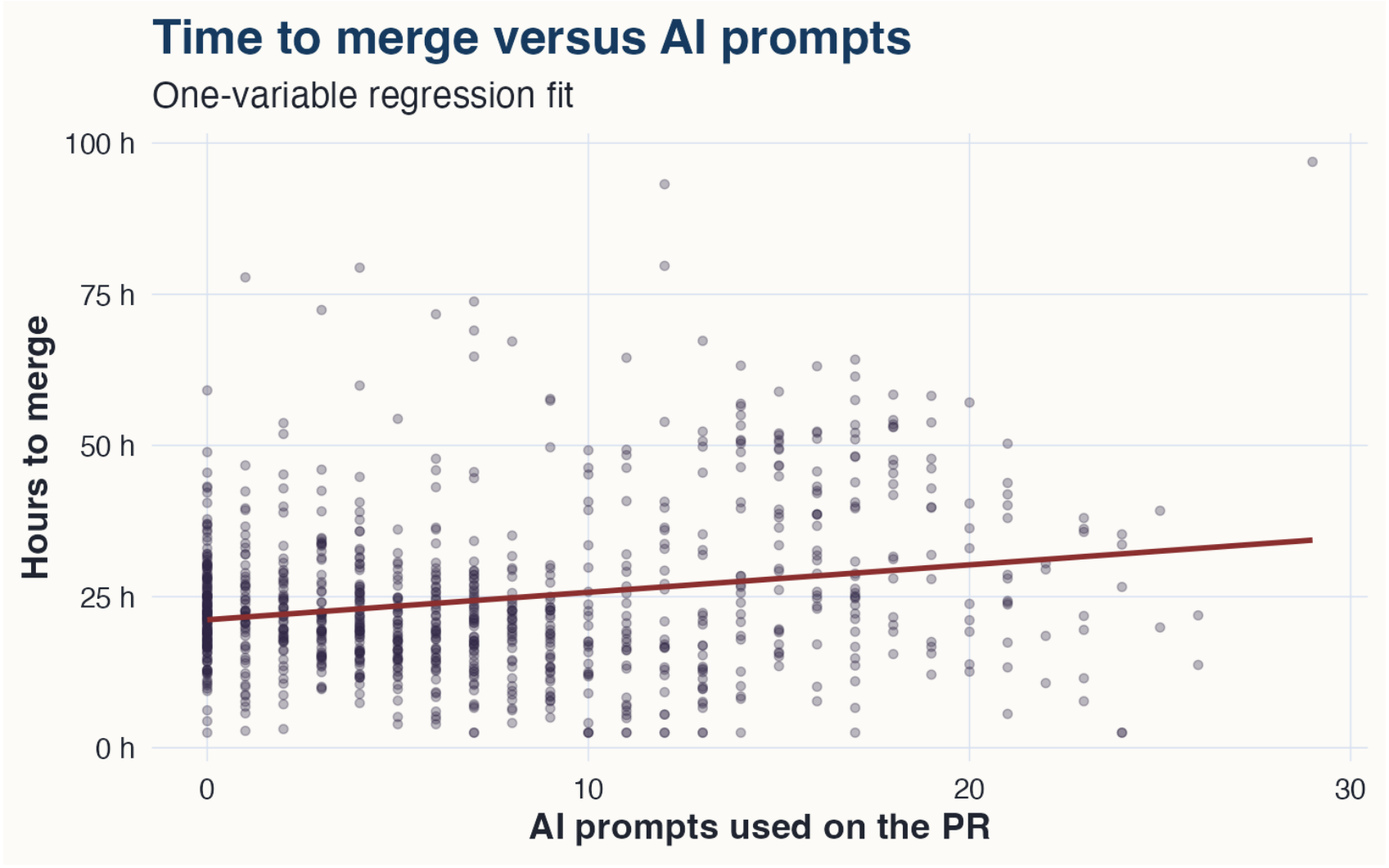
The question: **Did AI-assistant use speed up PRs?**

A few rows

```
prs %>% sample_n(8)
```

PR	AI prompts	lines changed	hours to merge
180	3	74	19.6
106	0	54	16.9
314	12	310	29.3
820	1	29	19.1
206	7	501	44.6
524	5	59	14.2
740	1	61	16.9
810	17	387	25.1

Already: these PRs are not all the same kind of task.


Time to merge versus AI prompts
One-variable regression fit
100 h
75 h
50 h
25 h
0 h
Hours to merge
0
10
20
30
AI prompts used on the PR

First read

The one-variable slope is positive.

Naive read:

PRs with more AI prompts took longer to merge.

So is the AI assistant slowing everyone down?

The naive model

Fit the model we already know:

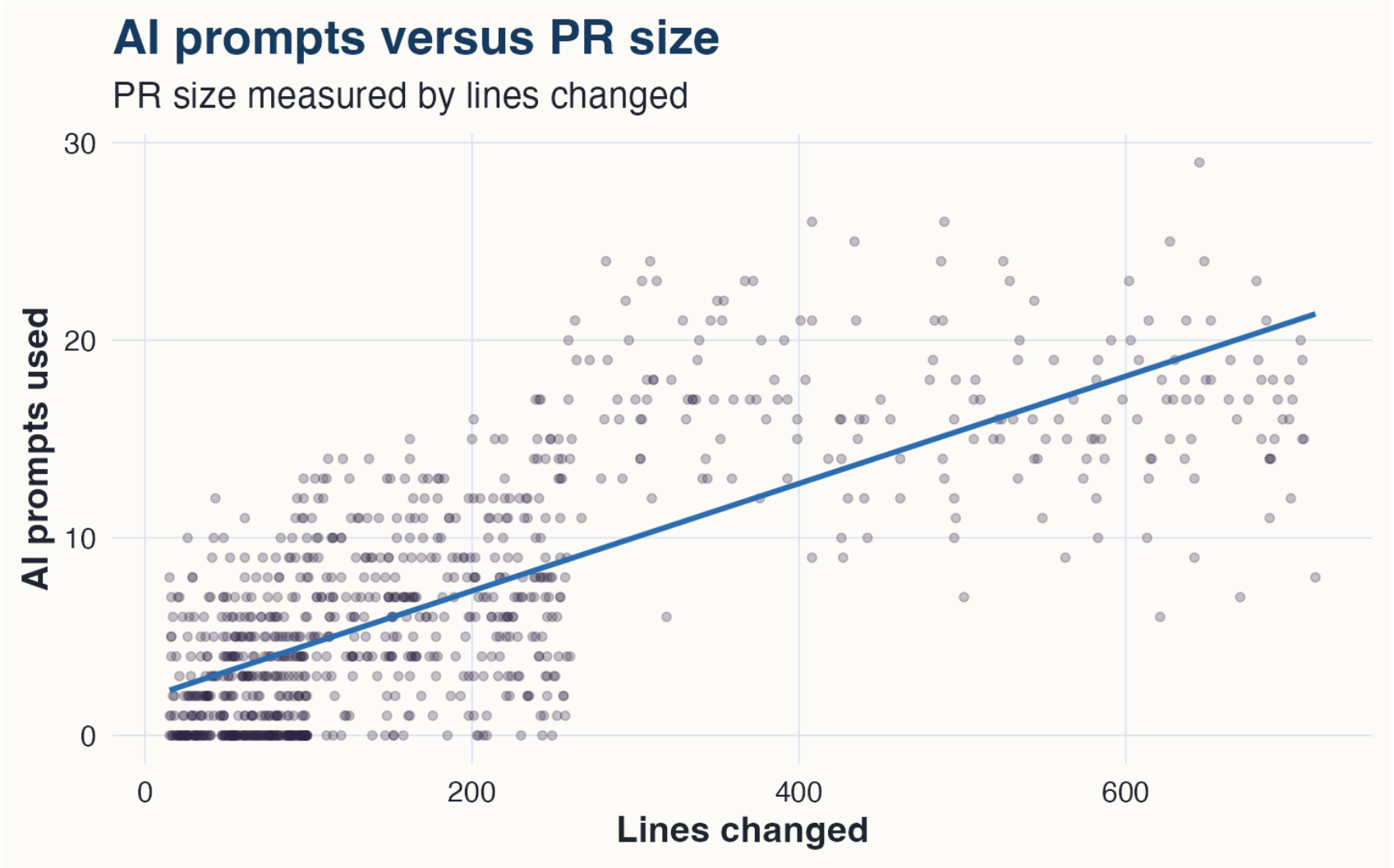
$$\text{HoursToMerge}_i = \beta_0 + \beta_1 \text{AIPrompts}_i + e_i.$$

For these data:

$$\widehat{\text{HoursToMerge}} = 21.2 + 0.45 \text{AIPrompts}.$$

Naive answer: each additional AI prompt predicts about **0.45 more hours** before merge.

That sounds bad.


AI prompts versus PR size
PR size measured by lines changed
AI prompts used
30
20
10
0
0
200
400
600
Lines changed

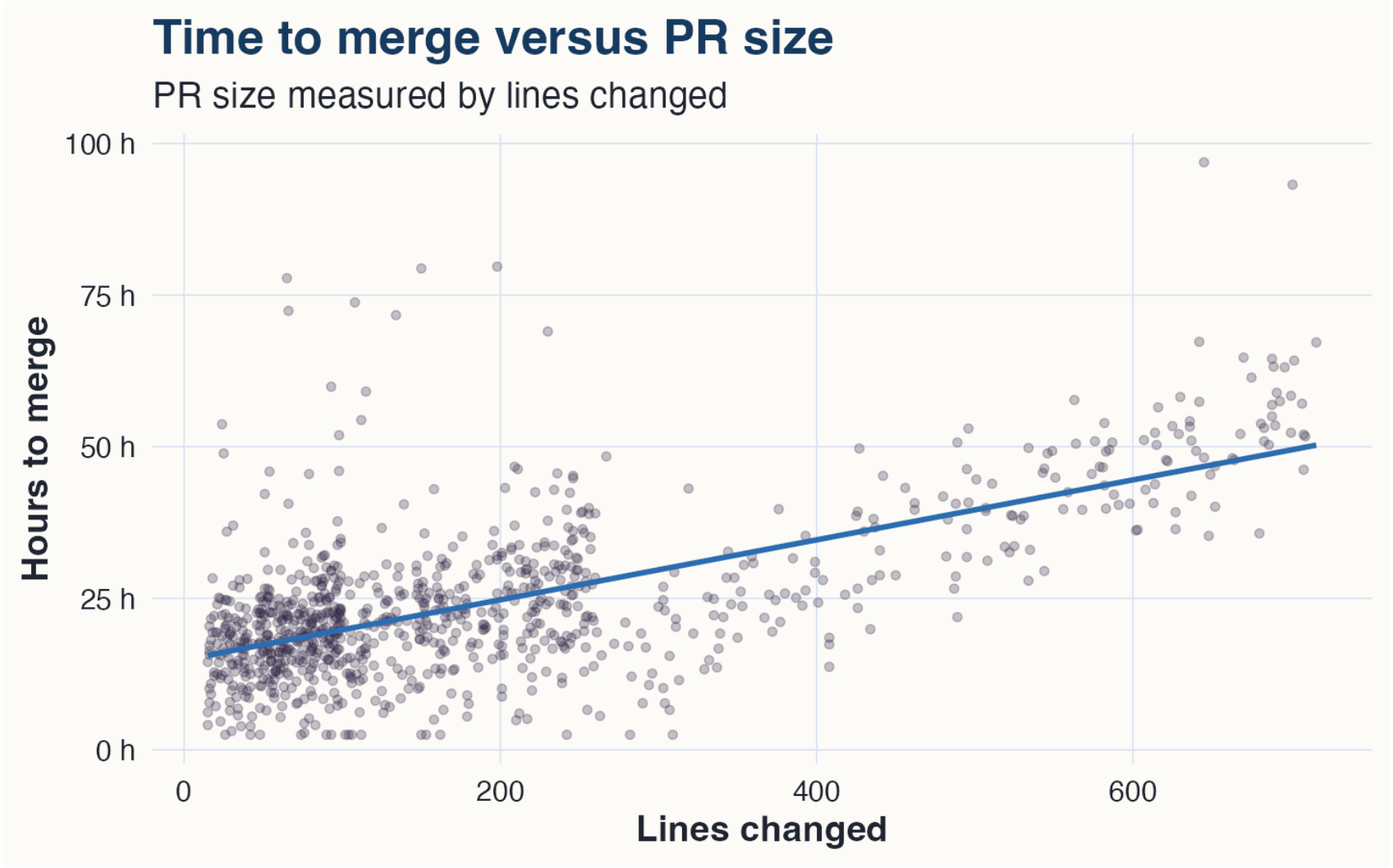
What the plot says

Bigger PRs use more AI prompts.

That is not surprising.

If a PR changes 40 lines, maybe you ask one question.

If it changes 800 lines, you might ask many.


Time to merge versus PR size
PR size measured by lines changed
100 h
75 h
50 h
25 h
0 h
Hours to merge
0
200
400
600
Lines changed

The confounding pattern

Bigger PRs also take longer to merge.

So `LinesChanged` is related to:

- the predictor we care about, `AIPrompts`,
- the outcome, `HoursToMerge`.

That is the confounding pattern again.

The wrong comparison

The naive model compares:

PRs with many prompts versus PRs with few prompts.

But because prompt count and PR size are related, it is also comparing:

larger PRs versus smaller PRs.

That is not the comparison we meant.

We want something closer to:

compare PRs of similar size.

A simple adjustment idea

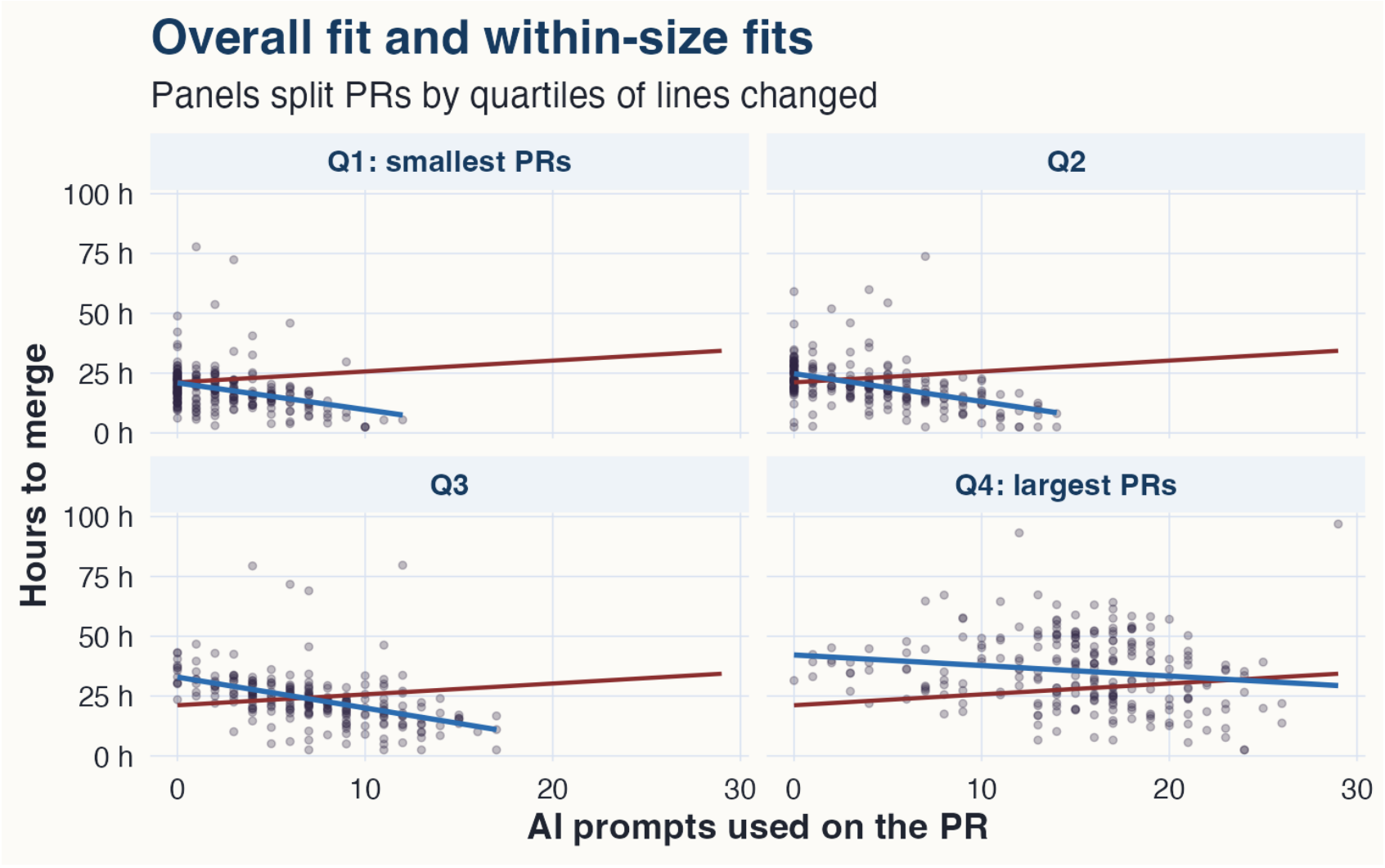
Split PRs into quartiles by `LinesChanged`.

Then compare prompt use **within each size range**.

```
prs = prs %>%
  mutate(SizeQuartile = cut_number(LinesChanged, 4))
```

This is not the final method.

It is a picture of the adjustment we want regression to do.


Overall fit and within-size fits
Panels split PRs by quartiles of lines changed
Q1: smallest PRs
Q2
Q3
Q4: largest PRs
Hours to merge
100 h
75 h
50 h
25 h
0 h
0
10
20
30
AI prompts used on the PR

Reading the quartile plot

Red line:
compare PRs of all sizes.

Blue lines:
compare PRs of similar size.

The sign changes.

What happened?

The overall relationship says:

> *PRs with more AI prompts take longer.*

The within-size relationship says:

> *among similarly sized PRs, more AI prompts predict faster merges.*

Those are not contradictory.

They are different comparisons.

From bins to a continuous adjustment

The quartile plot adjusted for size by making four bins.

That was useful because we could see the logic:

> *compare high-prompt and low-prompt PRs among similarly sized PRs.*

But `LinesChanged` is not really a four-level variable.

It is numerical.

So regression can do the same comparison more directly.

Keep size numerical

Instead of asking:

which size quartile is this PR in?

ask:

how many lines changed?

The model then uses all the size information, not just a bin label.

That is the move from stratified adjustment to multiple regression with two numerical inputs.

From lines to planes

With one numerical predictor:

$$y_i = \beta_0 + \beta_1 x_i + e_i.$$

With two numerical predictors:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + e_i.$$

The fitted object is no longer a line in a two-dimensional plot.

It is a plane in a three-dimensional cloud.

The multiple regression model

Use both predictors:

$$\text{HoursToMerge}_i = \beta_0 + \beta_1 \text{AIPrompts}_i + \beta_2 \text{Lines100}_i + e_i.$$

where

$$\text{Lines100}_i = \frac{\text{LinesChanged}_i}{100}.$$

The fitted equation

For these data:

$$\widehat{\text{HoursToMerge}} = 17.3 - 1.3\text{AIPrompts} + 8.5\text{Lines100}.$$

Interpretation:

- among PRs with similar size, one more AI prompt predicts about **1.3 fewer hours** to merge,
- among PRs with similar prompt counts, 100 more changed lines predicts about **8.5 more hours** to merge.

Partial relationships

The coefficient on `AIPrompts` is a **partial slope**.

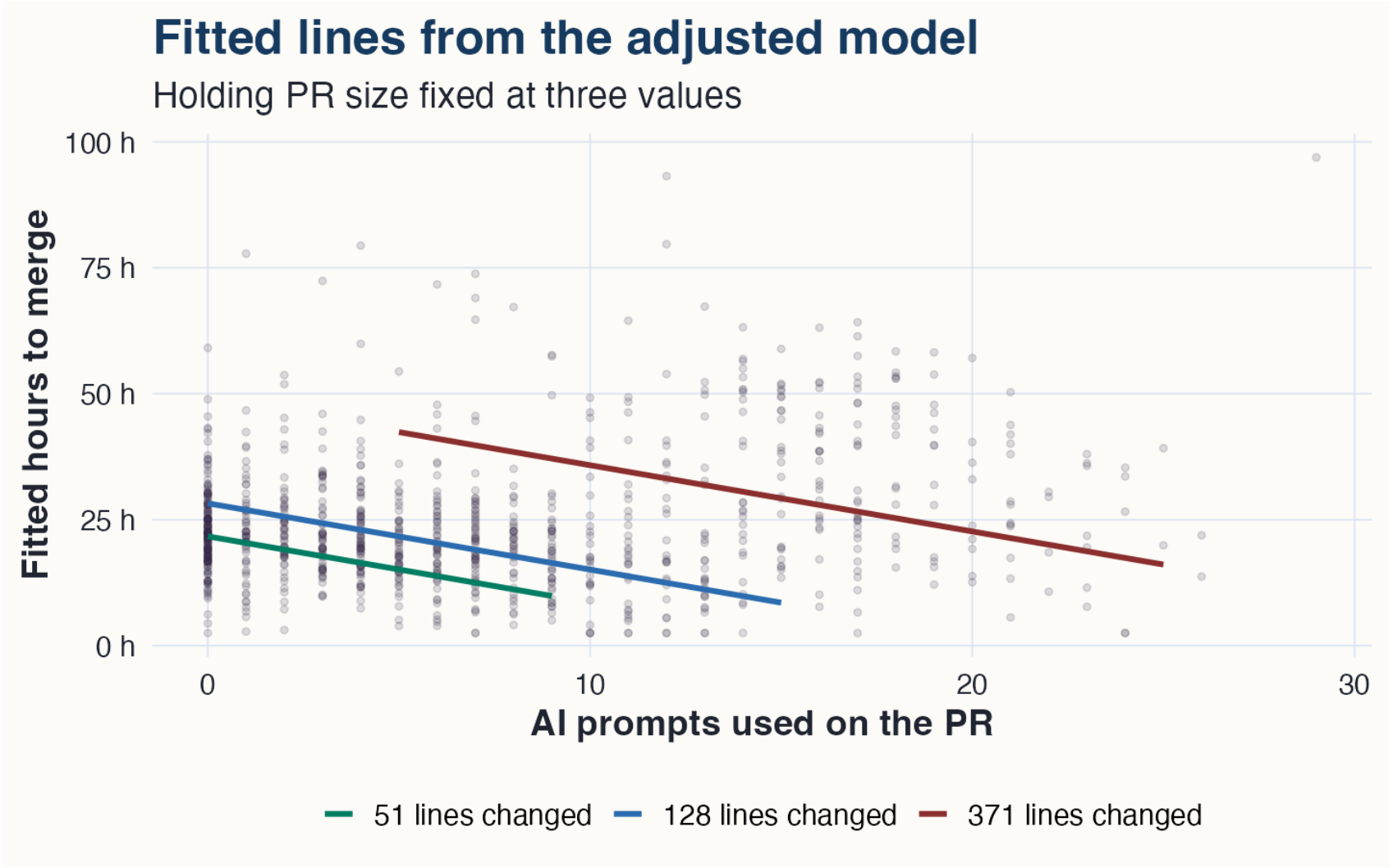
It answers:

> *How does merge time change with prompt count, holding PR size fixed?*

The one-variable slope was an **overall slope**.

It answered:

> *How do prompt count and merge time vary together, without holding PR size fixed?*


Fitted lines from the adjusted model
Holding PR size fixed at three values
Fitted hours to merge
100 h
75 h
50 h
25 h
0 h
0
10
20
30
AI prompts used on the PR
51 lines changed
128 lines changed
371 lines changed

Reading the fitted lines

This plot is a two-dimensional slice through the multiple regression model.

Each line holds PR size fixed at one value.

Same slope for prompts.

Different baselines for PR size.

Adjustment as subtraction

Start with the fitted equation:

$$\widehat{\text{HoursToMerge}} = 17.3 - 1.3\text{AIPrompts} + 8.5\text{Lines100}.$$

Move the PR-size part to the left:

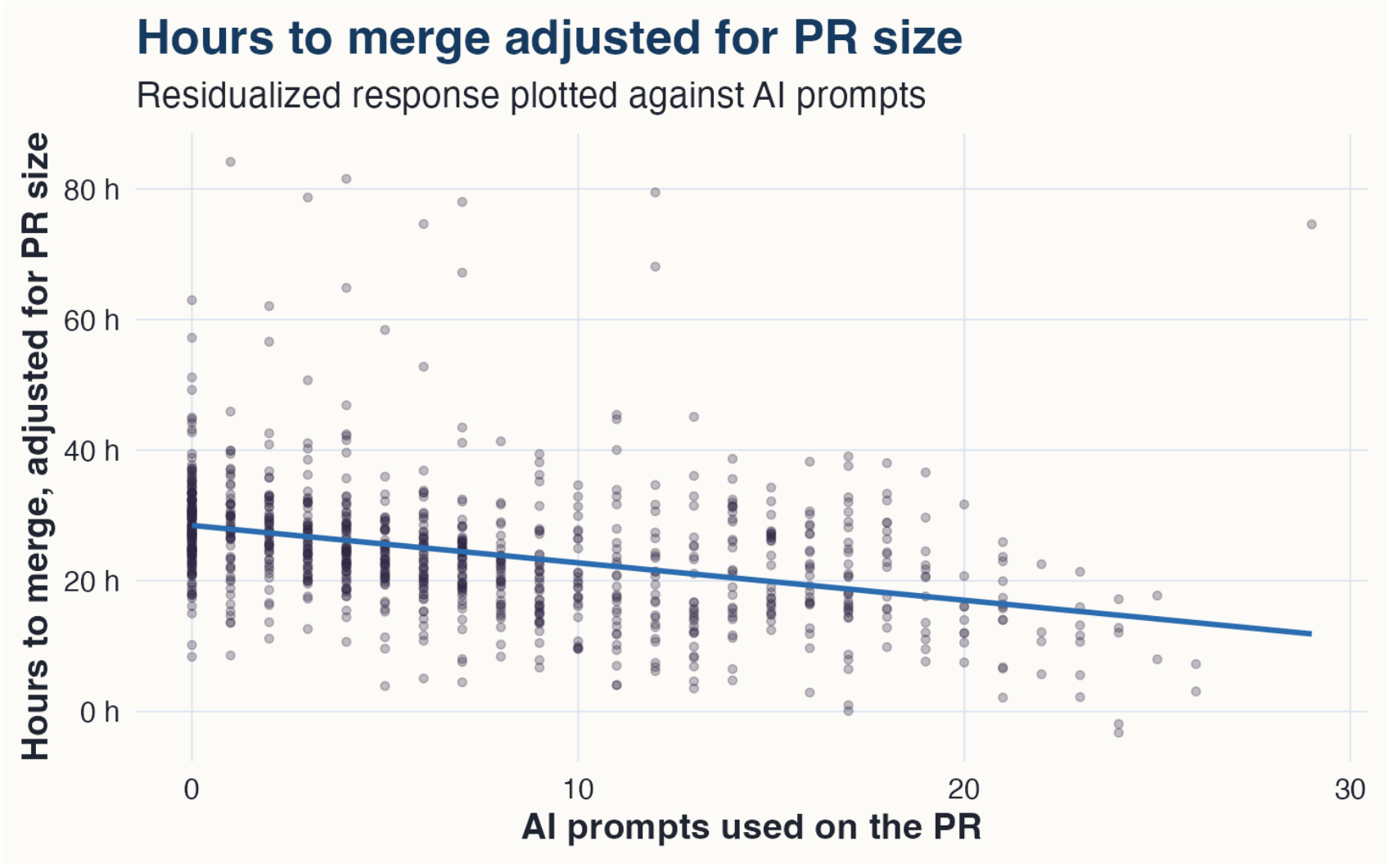
$$\widehat{\text{HoursToMerge}} - 8.5\text{Lines100} = 17.3 - 1.3\text{AIPrompts}.$$

Left side:

merge time, adjusted for PR size.

Right side:

the partial relationship with AI prompts.


Hours to merge adjusted for PR size
Residualized response plotted against AI prompts
Hours to merge, adjusted for PR size
80 h
60 h
40 h
20 h
0 h
0
10
20
30
AI prompts used on the PR

Reading the partial picture

After removing the PR-size pattern, the prompt relationship is negative.

This is the blue-line story from the quartile plots, compressed into one model.

What regression bought us

The quartile plot was useful because it made the problem visible.

But it was also clunky:

- quartile cutoffs are arbitrary,
- PRs near a cutoff get separated,
- we get four slopes instead of one clean adjusted estimate.

Multiple regression adjusts for `LinesChanged` directly.

The warning label

This is observational data.

So we should not announce:

> *AI prompts caused faster merges.*

Better:

> *After adjusting for PR size, PRs with more AI-assistant prompts tended to merge faster.*

That is still useful.

It is just not a randomized experiment.

Practical lesson

When predictors are correlated, a one-variable slope can answer the wrong question.

Before interpreting a slope, ask:

What comparison is this coefficient making?

Overall relationship:

compare everyone.

Partial relationship:

compare units that are similar on the adjustment variable.

A pricing question

H-E-B sells a weeknight meal kit through its grocery app.

- The meals change week to week: enchilada trays, salmon dinners, fajita kits, stir-fry bowls.
- Not a niche product. Lots of customers buy it because it solves an obvious problem.
- Customers first land on the meal-kit page.
- Conditional on that page view, some eligible viewers get a homepage or push promotion.

The business question: **if we raise price, how much demand do we lose?**

The app has some week-to-week price variation. But this was not a clean price experiment.

What got recorded

Each row is one week in a single Texas area, aggregated across stores within that area.

- `Market`: the Texas area.
- `UnitsSold`: number of kits sold,
- `Price`: average price for meal kits sold that week,
- `EligibleViews`: meal-kit page views eligible for the marketing push,
- `Marketing`: whether eligible viewers got the homepage/push promotion,

We start with `log(UnitsSold)` versus `log(Price)`.

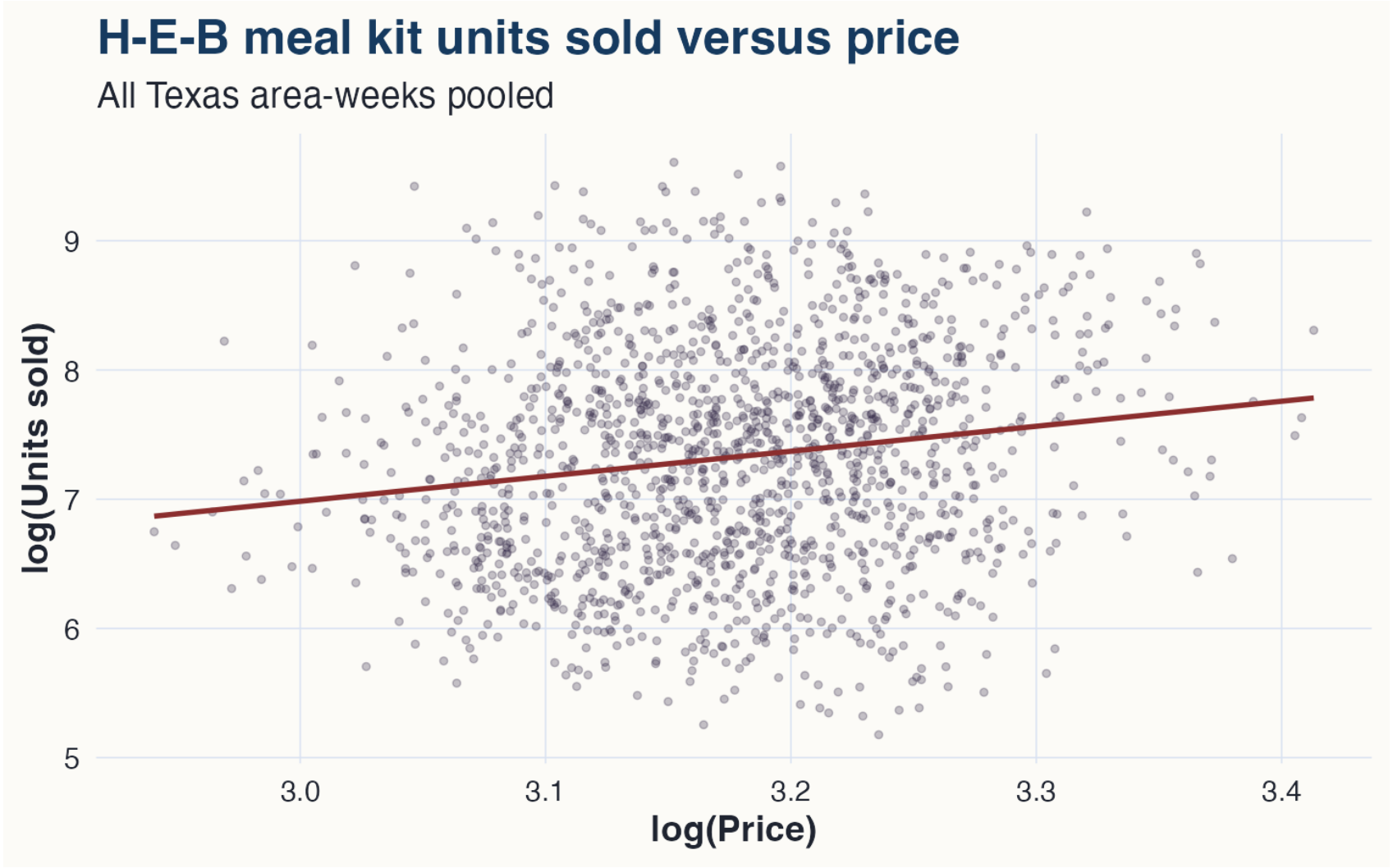
- Very common in econ-adjacent problems.
- In a log-log regression, the price coefficient is an **elasticity**.
- But units sold also depend on how many shoppers reached the meal-kit page.

A few rows

```
meal_kits %>% sample_n(8)
```

Market	Week	Price	Eligible views	Marketing	Units sold
East Texas	64	23.80	10749	Push	1025
Houston Central	44	24.43	32608	Push	10978
Rio Grande Valley	22	21.47	9179	Push	1258
San Antonio South	80	21.72	11891	Push	547
Austin North	52	23.23	16339	Push	3307
Houston Central	21	28.44	35809	No push	5089
Houston West	56	25.58	21210	Push	4016
Frisco-Plano	81	25.70	19581	Push	4427

Prices are around ordinary dinner for two money


H-E-B meal kit units sold versus price
All Texas area-weeks pooled
log(Units sold)
9
8
7
6
5
3.0
3.1
3.2
3.3
3.4
log(Price)

First read

The pooled relationship is positive.

That cannot be the demand curve we meant to estimate.

The naive model

Fit the one-variable model:

$$\log(\text{UnitsSold}_i) = \beta_0 + \beta_1 \log(\text{Price}_i) + e_i.$$

For these data:

$$\widehat{\log(\text{UnitsSold})} = 1.18 + 1.94 \log(\text{Price}).$$

Naive answer: higher prices predict more sales.

That is not a pricing answer. It is a warning about the comparison.

But marketing is not random

Marketing pushes likely increase demand, but also appear more often in lower-price weeks.

Marketing	Weeks	Average price	Average eligible views	Average units
No push	879	24.17	17368	1883
Push	681	23.84	16093	2615

So price is confounded with the marketing push.

Adjust for marketing

Add a marketing indicator:

$$\log(\mathsf{UnitsSold}_i) = \beta_0 + \beta_1 \log(\mathsf{Price}_i) + \beta_2 \mathsf{Push}_i + e_i.$$

Fitted equation:

$$\log(\widehat{\mathsf{UnitsSold}}) = 0.55 + 2.09 \log(\mathsf{Price}) + 0.28 \mathsf{Push}.$$

The price slope is still positive.

Marketing adjustment alone has not fixed the comparison.

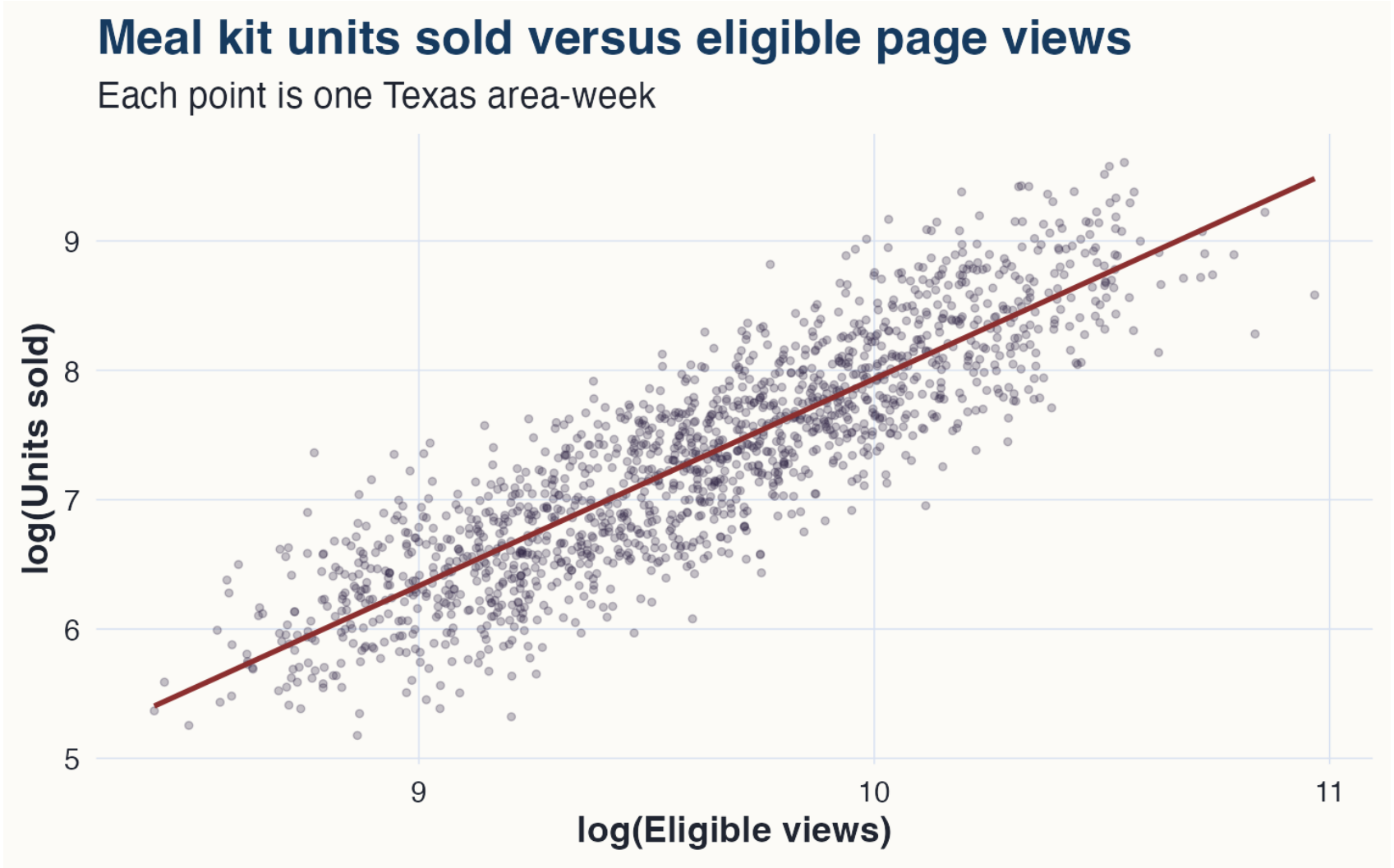
Exposure matters

`EligibleViews` is measured before the marketing push.

The sequence is:

1. a shopper reaches the meal-kit page,
2. the app may show a push to eligible viewers,
3. the shopper may buy.

So `EligibleViews` is an exposure variable. It tells us how many shoppers had a chance to buy.


Meal kit units sold versus eligible page views
Each point is one Texas area-week
log(Units sold)
9
8
7
6
5
9
10
11
log(Eligible views)

Adjust for exposure

Now include both numerical predictors:

$$\log(\text{UnitsSold}_i) = \beta_0 + \beta_1 \log(\text{Price}_i) + \beta_2 \log(\text{EligibleViews}_i) + \beta_3 \text{Push}_i + e$$

For these data:

$$\hat{\beta}_1 = -2.71, \qquad \hat{\beta}_2 = 1.81.$$

Price now slopes downward. Eligible views are doing what they should: more eligible viewers means more sales.

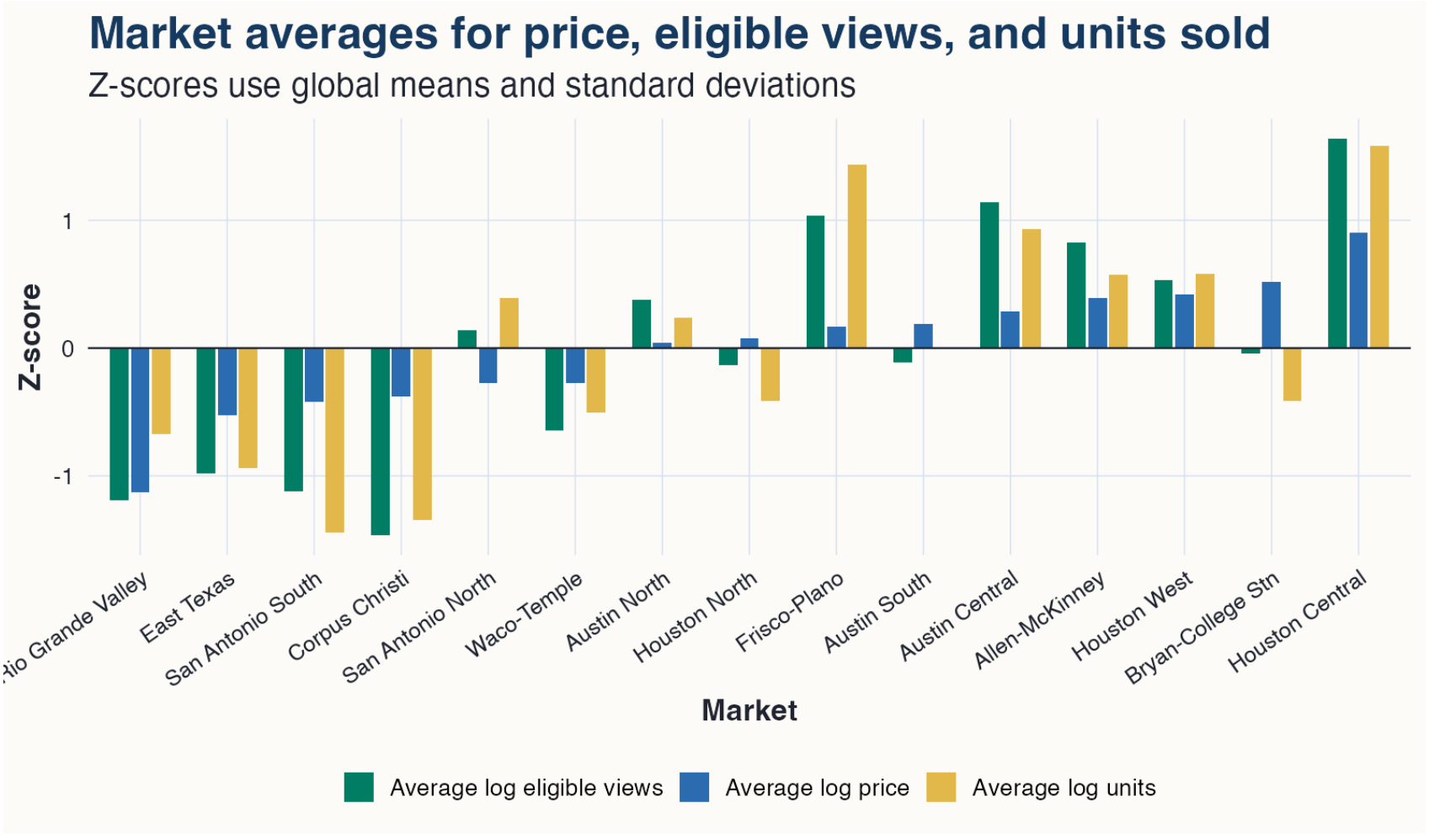
Something else is moving

Prices, eligible views, and demand all vary a lot by market.

That is normal:

- markets differ in size,
- markets differ in income and delivery costs,
- local teams may choose different price levels,
- H-E-B is not equally mature in every Texas area.

So a pooled regression may be comparing Houston Central weeks to Rio Grande Valley weeks, not just expensive weeks to cheap weeks.


Market averages for price, eligible views, and units sold
Z-scores use global means and standard deviations
Z-score
1
0
-1
Rio Grande Valley
East Texas
San Antonio South
Corpus Christi
San Antonio North
Waco-Temple
Austin North
Houston North
Frisco-Plano
Austin South
Austin Central
Allen-McKinney
Houston West
Bryan-College Stn
Houston Central
Market
Average log eligible views
Average log price
Average log units

What market averages show

Some markets are high-price, high-exposure, and high-demand.

Adjust for market

Now compare weeks in the same market:

$$\log(\text{UnitsSold}_i) = \beta_0 + \beta_1 \log(\text{Price}_i) + \beta_2 \log(\text{EligibleViews}_i) + \beta_3 \text{Push}_i + \gamma$$

The fitted price slope is:

$$\hat{\beta}_1 = -2.35.$$

Interpretation:

comparing weeks in the same area, with similar eligible page views and the same marketing status, a 1% price increase predicts about 2.35% fewer units sold.

Revenue implication

For a log-log demand model:

$$\text{revenue elasticity} \approx 1 + \text{demand elasticity}.$$

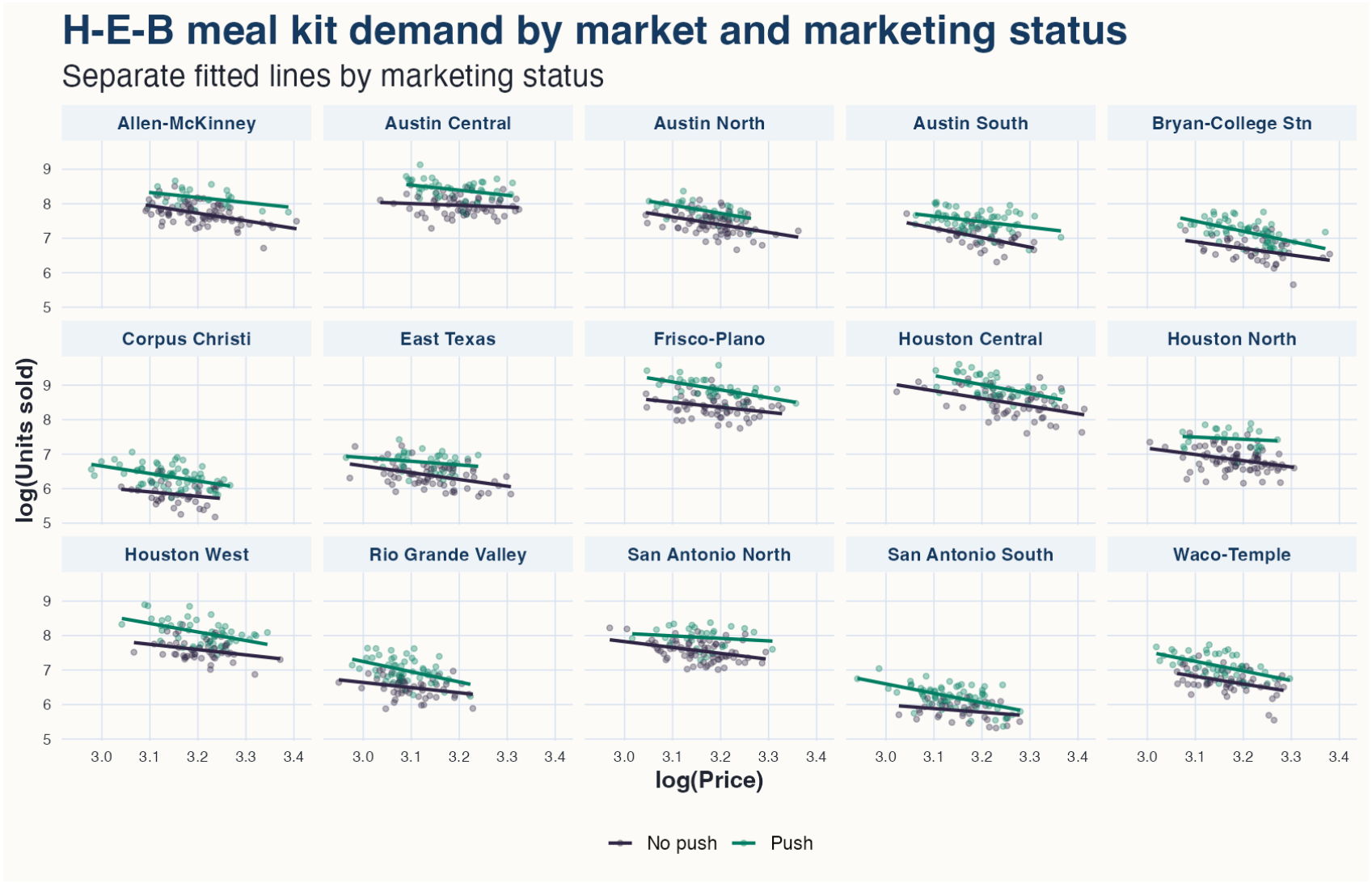
With the market-adjusted estimate:

$$1 + (-2.35) = -1.35.$$

So a 1% price increase predicts roughly:

- 2.35% fewer units,
- 1.35% lower revenue.

That is a very different business answer.


H-E-B meal kit demand by market and marketing status
Separate fitted lines by marketing status
Allen-McKinney
Austin Central
Austin North
Austin South
Bryan-College Stn
Corpus Christi
East Texas
Frisco-Plano
Houston Central
Houston North
Houston West
Rio Grande Valley
San Antonio North
San Antonio South
Waco-Temple
log(Units sold)
log(Price)
No push
Push

Reading the market fits

Within markets, push weeks tend to sit higher.

The push-week slopes also look a bit steeper.

So one more question is natural:

> *Is the price elasticity the same in push weeks and no-push weeks?*

Add the interaction

Let the price slope differ by marketing status.

Remember: mean-center numerical predictors when adding interactions. Define:

$$x_i = \log(\text{Price}_i) - \log(23).$$

Then fit:

$$\log(\text{UnitsSold}_i) = \beta_0 + \beta_1 x_i + \beta_2 \text{Push}_i + \beta_3 x_i \text{Push}_i + \beta_4 \log(\text{EligibleViews}_i)$$

The centering makes the push coefficient refer to a price of about \$23.

The fitted interaction

For the interaction model:

$$\hat{\beta}_1 = -2.23, \qquad \hat{\beta}_3 = -0.28.$$

So:

Marketing status	Estimated price elasticity
No push	-2.23
Push	-2.51

At a price of about $23, push weeks have about **55% higher** expected unit sales, after adjusting for eligible views and market.

What might have happened?

The data do not look like a fully randomized pricing experiment.

A reasonable diagnosis:

- local price tests created week-to-week price variation,
- marketing pushes were more common in cheaper weeks,
- the number of eligible page views varied a lot by area and week,
- some Texas areas had higher prices and higher baseline demand,
- pooling all markets mixed those patterns together.

The first slope was the right answer to the wrong question.

Practical lesson, again

For the price question, the useful comparison is:

> *higher-price weeks versus lower-price weeks, within the same market and marketing condition.*

The final model estimates that comparison.

That is the move from **overall relationship** to **partial relationship, holding relevant factors constant.**